# BEACON – Disaster Relief Resource Management System

**A MINI PROJECT REPORT**

*Submitted in partial fulfilment of the requirements of*

**EVENT DRIVEN & UI PROGRAMMING**

for

**B.TECH. INTEGRATED**
**(COMPUTER ENGINEERING)**
Year III – Semester VI – Div. C

*Academic Year 2025 - 2026*

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## 1. Introduction & Problem Statement

Natural disasters such as floods, earthquakes, cyclones, and landslides are increasing in frequency and impact. In a country like India, where population density is high, even a single disaster can affect thousands or even millions of people within a short time.

One of the biggest challenges during such situations is not just the lack of resources, but the mismanagement and uneven distribution of available resources. In many cases:

- Some areas receive excess help, while others are ignored
- Donors are willing to help but don't know what is actually needed
- NGOs and shelters struggle to communicate their real-time requirements
- There is no proper system to track whether help has reached the affected people

Currently, information is scattered across news channels, social media, and local authorities, which makes it difficult to act quickly and efficiently.

To address these issues, we designed BEACON, a UI-based platform that acts as a centralized system for disaster relief coordination. The platform allows users to:

- Identify disaster-affected areas using a visual map
- Report disasters quickly
- View and understand real-time resource requirements
- Donate resources or money
- Track donation progress

The main objective of this project is to create a simple, user-friendly interface that improves communication and coordination during disaster situations.

## 2. Individual Contribution(s)

Vihaan Patole (C092)

- Worked on map interface and disaster visualization screens
- Contributed to reporting page design
- Assisted in improving interaction with map-based features

Darren D'silva (C108)

- Worked on preparing the project report and structuring the content
- Contributed to defining the overall workflow and feature explanation
- Assisted in reviewing UI screens and improving clarity in navigation

Shreya Sanghvi (C113)

- Worked on creating the project presentation (PPT)
- Organized content into clear and understandable slides
- Assisted in aligning presentation with the UI design

Abdul Sheikh (C180)

- Worked on designing UI screens in Figma
- Contributed to layouts for donation and tracking features
- Helped maintain consistency in design elements

Anuradha Bansal (C181)

- Worked on NGO/Shelter and requirements pages
- Helped organize resource categories and content layout
- Assisted in maintaining overall UI consistency

## 3. User Research & Problem Analysis

Before designing the UI, we analyzed the different types of users who would interact with the system.

### Target Users:

- **Affected Individuals**
  - People directly impacted by disasters
  - Need immediate help such as food, shelter, and medical support
- **Donors**
  - Individuals or organizations willing to contribute
  - Need clarity on what to donate and where
- **NGOs and Shelters**
  - Organizations managing relief operations
  - Need a platform to communicate requirements and receive support

### Problems Identified:

- **Lack of Centralized Information**
  - No single platform to view all affected areas
  - Users rely on scattered sources
- **Unclear Requirements**
  - Donors often don't know what is needed
  - Leads to wastage or irrelevant donations
- **Trust Issues**
  - Donors hesitate due to lack of verification
  - Fear of fraud or misuse
- **Poor Tracking System**
  - No visibility on whether donations are delivered
  - Lack of transparency

### Key Observations:

- Users prefer quick actions instead of complex forms
- Visual representation (like maps) improves understanding
- Simple UI increases usability during stressful situations
- Trust indicators (like verified NGOs) are important

These insights were used to guide the design of the platform.

## 4. Proposed Solution

The proposed solution is BEACON, a disaster relief management platform that connects all stakeholders through a single interface.

The system focuses on three main aspects:

- Identification of affected areas
- Communication of requirements
- Facilitation of donations

### Working of the System:

1. **User Authentication**
    - Users can sign up or log in to access features
2. **Home Screen with Map**
    - Displays affected regions using color coding
    - Helps users understand severity
3. **Reporting System**
    - Users can report disasters by selecting location and type
    - Can also mention required resources
4. **Requirements Page**
    - Displays needs such as food, clothes, or funds
    - Helps donors make informed decisions
5. **Donation System**
    - Users can contribute money or resources
    - Simple and guided process
6. **NGO/Shelter Page**
    - Shows verified organizations
    - Provides additional details
7. **Tracking System**
    - Displays donation status (Paid → In Transit → Delivered)

Why This Solution Works:

- Centralizes information
- Reduces confusion
- Improves transparency
- Encourages more people to contribute

## 5. Design Process

### Step 1: Understanding the Problem

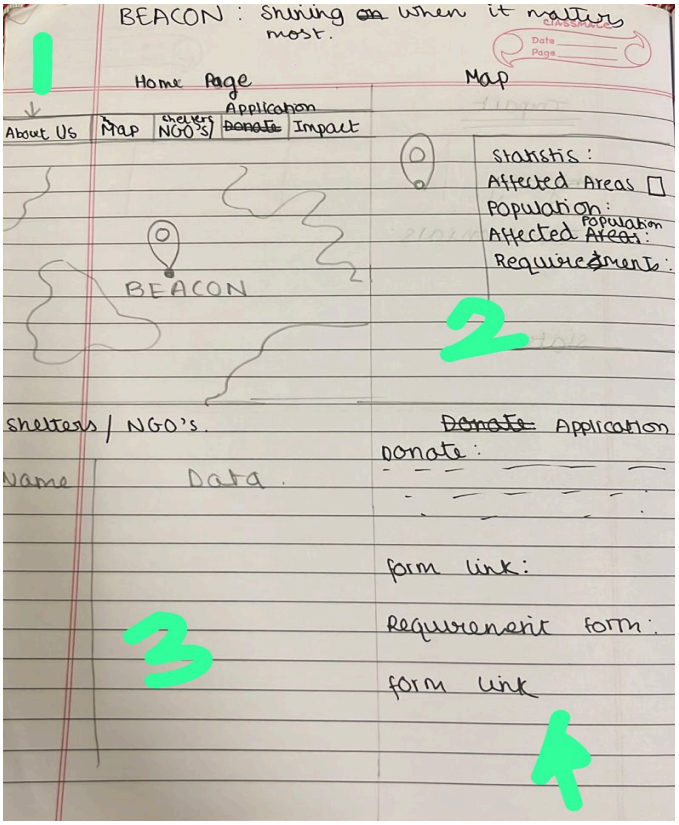
We started by identifying the main issues in disaster management systems. The focus was on speed, clarity, and accessibility.

### Step 2: Sketching & Wireframing

Basic sketches were created to:

- Plan layout of screens
- Decide placement of buttons and elements
- Keep navigation simple

These wireframes helped in visualizing the structure before designing.



**Step 3: UI Design in Figma**

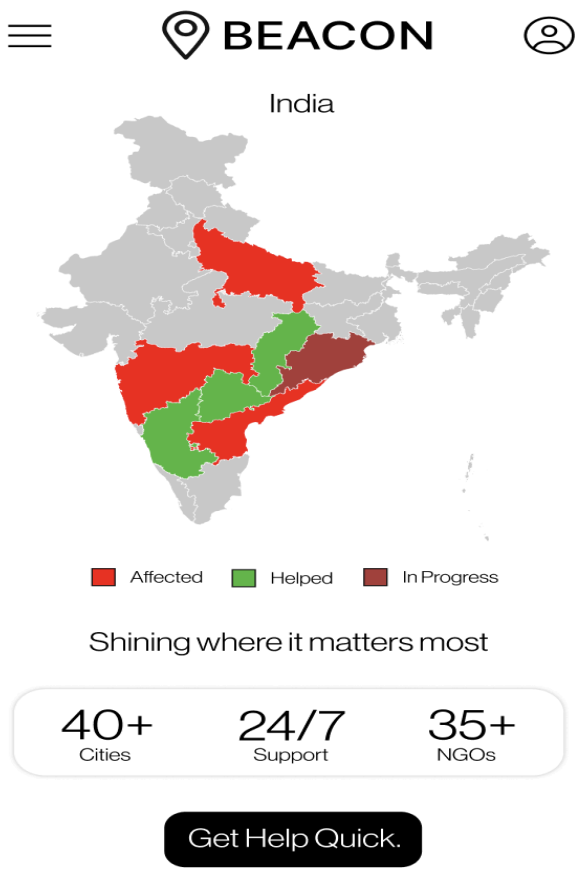
The final UI was designed using Figma.

Design choices included:

- Minimal color palette
- Clear typography
- Use of icons for better understanding
- Consistent layout across screens

Special attention was given to:

- Button placement
- Readability



**Step 4: Prototyping**

The screens were connected to create a working prototype. This helped in:

- Testing navigation
- Identifying usability issues
- Improving flow

**Step 5: Iteration**

Based on feedback and testing:

- Some layouts were simplified
- Navigation was improved
- Content was made more clear

## 6. Key Features

### 1. Disaster Reporting Feature

Users can report disasters easily by:

- Selecting location
- Choosing disaster type
- Adding requirements

This ensures quick communication of issues.

### 2. Interactive Map

- Displays affected areas visually
- Uses color coding for clarity
- Helps users understand the situation quickly

### 3. Requirements Display

- Categorized into:
    - Food
    - Clothes
    - Money
- Helps donors decide what to contribute

### 4. Donation System

- Allows easy contribution
- Shows breakdown of amount
- Simple and user-friendly

### 5. NGO/Shelter Listing

- Displays verified organizations
- Builds trust
- Helps users connect directly

### 6. Donation Tracking

- Shows real-time status:
    - Paid
    - In Transit
    - Delivered
- Ensures transparency

## 7. Conclusion

The BEACON platform presents a practical and user-friendly approach to managing disaster relief resources through a centralized system. The main strength of this project lies in its ability to bring together different users—affected individuals, donors, and NGOs—on a single platform and simplify the entire process of disaster response.

Through this project, we understood that during disasters, the issue is not always the lack of resources, but the lack of proper communication and coordination. By introducing features such as disaster reporting, map-based visualization, requirement listing, and donation tracking, the system helps reduce confusion and ensures that support reaches the right places.

The use of a clean and simple UI plays an important role, especially in emergency situations where users may be under stress. The design focuses on quick actions, minimal steps, and clear information, making it easy for users to navigate and perform tasks like reporting or donating without difficulty.

Another important aspect of the system is transparency. The donation tracking feature and display of verified NGOs help in building trust among users, which is often a major concern in real-world scenarios.

While the current project is a UI prototype, it has strong potential for future development. With backend integration, real-time data, and location services, BEACON can be expanded into a fully functional system that can assist during actual disaster situations.

Overall, this project helped us understand the importance of user-centered design and how technology can be used to solve real-world problems effectively. It also gave us practical experience in designing interfaces that are not only visually appealing but also meaningful and purposeful.

## 8. References

*Links*


- https://www.figma.com/
- https://developers.google.com/maps
- https://ndma.gov.in/ (National Disaster Management Authority, India)
- https://www.redcross.org/
- https://www.un.org/en/global-issues/disaster-risk-reduction
- https://reliefweb.int/


*Books*


- *"Don't Make Me Think" – Steve Krug*
  *(Basic usability and UI principles)*
- *"The Design of Everyday Things" – Don Norman*
  *(Understanding user-centered design and interaction)*
- *"Designing Interfaces" – Jenifer Tidwell*
  *(Patterns and best practices for UI design)*
- *"Interaction Design: Beyond Human-Computer Interaction" – Preece, Rogers, Sharp*
  *(Concepts of interaction design and user experience)*